

SANJIV RANCHOD

CURRICULUM VITAE

PERSONAL INFORMATION

<i>Date of Birth</i>	16 September 1998
<i>Nationality</i>	South African
<i>Phone</i>	+32 479 226866
<i>Email</i>	ranchodsanjiv@gmail.com
<i>Academic Webpage</i>	sanjivranchod.com
<i>Address</i>	37 Northampton Street Cambridge CB3 0AD United Kingdom

RESEARCH EXPERIENCE & EDUCATION

PhD	<i>2022 - present</i>
Computer Science University of Cambridge (United Kingdom) Thesis: <i>Substitution for Substructural Theories</i>	
Master of Science	<i>2021 - 2022</i>
Mathematics (by dissertation) University of Cape Town (South Africa) Thesis: <i>Internal Factorisation Systems</i> <i>Degree Awarded with Distinction</i>	
Master of Science Exchange Student	<i>2020, 2022</i>
Mathematics Université Catholique de Louvain (Belgium) Thesis: <i>Internal Categories and Factorisation Systems</i>	
Bachelor of Science Honours	<i>2020</i>
Mathematics University of Cape Town (South Africa) GPA: 92 Thesis: <i>Torsion Theories</i> <i>Degree Awarded in the First Class</i>	
Bachelor of Science	<i>2017 - 2019</i>
Mathematics and Applied Mathematics University of Cape Town (South Africa) GPA: 93.29 Thesis: <i>Internal Categories</i> <i>Degree Awarded with Distinction</i>	
High School	<i>2012 - 2016</i>
Westerford High School (South Africa) <i>9 distinctions</i>	

PUBLICATIONS

Substitution for Varieties of Substructural Theories (<i>with Marcelo Fiore</i>)	<i>2026</i>
Submitted	
Substructural Abstract Syntax with Variable Binding and Single-Variable Substitution (<i>with Marcelo Fiore</i>)	<i>2025</i>
Proceedings of Fortieth Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)	
A Finite Algebraic Presentation of Lawvere Theories in the Object-Classifier	<i>2025</i>
Topos <i>with Marcelo Fiore</i>	
Theory and Applications of Categories (TAC)	

AWARDS & FUNDING

Department of Computer Science and Technology Travel Grant (~R75 000)	2025
St John's College Learning and Research Fund (~R50 000)	2022
Gates Cambridge Scholarship (~R4 000 000)	2022
Harry Crossley Research Travel Award (~R10 000)	2022
Manuel & Luby Washkansky Scholarship (R15 450)	2021
Vice-Chancellor's Research Scholarship (R5 000)	2021
Harry Crossley Research Fellowship (R200 000)	2021
Erasmus+ International Learning Mobility Grant (~R100 000 each)	2020, 2022
UCT Honours Merit Award (R20 000)	2020
Council Scholarship (R3000)	2020
NRF Honours Innovation Scholarship (R60 000)	2020
MAM3040W Class Medal	2019
Science Faculty Scholarship	2018
Vice-Chancellor's Scholarship (R40 000 each)	2017, 2018, 2019
Dean's Merit List	2017, 2018, 2019

TEACHING

Postgraduate Demonstrator (University of Cambridge)	2023 - present
<i>Weekly exercise classes of ~15 students</i>	
Computer Science Part III/ACS: Category Theory (master's course)	
Undergraduate Supervisor (University of Cambridge)	2022 - present
<i>Weekly exercise classes of 1-3 students</i>	
Computer Science Part IA: Discrete Mathematics (first-year undergraduate)	
<i>King's College, Christ's College, Clare College, Emmanuel College, Selwyn College</i>	
Assistant Lecturer (University of Cape Town)	2021
<i>Weekly online lectures of ~50 students</i>	
MAM1005H: Introduction to Calculus (first-year undergraduate)	
Workshop Lecturer (University of Cape Town)	2018 - 2021
<i>Lectures of 10-150 students</i>	
Proof-writing workshop (undergraduate)	2019
Review lectures for mathematics courses (undergraduate)	2018 - 2022
<i>Supported by University Capacity Development Grant of Claire Blackman and Jonathan Shock</i>	
Undergraduate Tutor (University of Cape Town)	2018 - 2022
<i>Weekly exercise classes of 20-100 students</i>	
MAM1005H Bootcamp: Introduction to Calculus (first-year undergraduate)	2022
MAMBridge: Introduction to University Mathematics (first-year undergraduate)	2021
MAM3000W 3TA: Topics in Algebra (third-year undergraduate)	2021
MAM2000W 2AC: Advanced Calculus (second-year undergraduate)	2021
MAM2000W 2RA: Real Analysis (second-year undergraduate)	2020
MAM2000W 2IA: Introduction to Algebra (second-year undergraduate)	2019
MAM1019H: Fundamentals of Mathematics (undergraduate)	2018, 2019, 2020, 2021
MAM1000W: Introduction to Calculus (first-year undergraduate)	2018, 2019
Private High School and University Tutor	2017 - present
<i>One-on-one lessons</i>	
I tutor high school students in Mathematics, Physics and Advanced Programme Mathematics for Grades 7-12, and undergraduate and master's students in mathematics courses for Science, Engineering and Commerce programmes.	

TALKS & PRESENTATIONS

Single-Variable Substitution for Substructural Theories	2025
LIX Seminar, École Polytechnique	
<i>Invited Seminar Talk</i>	
Single-Variable Substitution for Substructural Theories	2025
GT Sémantique, IRIF, Université Paris Cité	
<i>Invited Seminar Talk</i>	

Substitution for Substructural Theories Annual International Category Theory Conference (CT), Masaryk University <i>Contributed Conference Talk</i>	2025
Substructural Abstract Syntax with Variable Binding and Single-Variable Substitution Logic in Computer Science (LICS), National University of Singapore <i>Contributed Conference Talk</i>	2025
Substitution for Linear-Cartesian and Full Substructural Theories Italian Category Theory Seminar (ItaCa), Online <i>Invited Seminar Talk</i>	2025
Category Theoretic Models of Substitution for Substructural Syntax Stockholm Logic Seminar, University of Stockholm <i>Invited Seminar Talk</i>	2025
Mixed Linear-Cartesian Theories as Substitution Monoids SANDWICH Seminar, University of Cambridge <i>Seminar Talk</i>	2024
Lawvere Theories and Symmetric Operads as Substitution Algebras: Free constructions for Abstract Syntax Annual International Category Theory Conference (CT), University of Santiago de Compostela <i>Contributed Conference Talk</i>	2024
A New Approach to Abstract Syntax with Variable Binding for Cartesian Theories Junior Category Theory Seminar, Université Catholique de Louvain <i>Invited Seminar Talk</i>	2024
Internal Factorisation Systems Topology, Algebra and Categories in Logic (TACL), University of Coimbra <i>Contributed Conference Talk</i>	2022
Internal Factorisation Systems South African Mathematical Society Congress, University of Cape Town <i>Contributed Conference Talk</i>	2021
Internal Categories South African Mathematical Society Congress, University of Cape Town <i>Contributed Conference Talk</i>	2019
What is a Category and Why Should You Care? UCT Mathematical Society, University of Cape Town <i>Invited Seminar Talk</i>	2019
OTHER ACADEMIC ACTIVITIES	
Organiser of SANDWICH Day Workshop Department of Computer Science and Technology University of Cambridge	2026
Organiser of CT2026 UK Satellite Event Department of Computer Science and Technology University of Cambridge	2026
Reviewer for Mathematical Structures in Computer Science <i>Published by Cambridge University Press</i>	2025
Organiser of the Logic and Semantics Seminar Department of Computer Science and Technology University of Cambridge	2025 - present
Organiser of the Topos Theory Reading Group Department of Computer Science and Technology University of Cambridge	2025 - present

Organiser of the SANDWICH Seminar
Department of Computer Science and Technology
University of Cambridge

2025 - present

TRAINING

Academic Teaching Internship Programme
Department of Mathematics and Applied Mathematics
University of Cape Town

2021

REFERENCES

Professor Marcelo Fiore
Department of Computer Science and Technology; Christ's College
University of Cambridge
marcelo.fiore@cl.cam.ac.uk

Professor George Janelidze
Department of Mathematics and Applied Mathematics
University of Cape Town
george.janelidze@uct.ac.za

Associate Professor Jonathan Shock
Department of Mathematics and Applied Mathematics
University of Cape Town
jonathan.shock@uct.ac.za

Professor Marino Gran
Institute for Research in Mathematics and Physics
Université Catholique de Louvain
marino.gran@uclouvain.be